



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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Department of COMPUTER SCIENCE AND ENGINEERING

Project Work Phase – I

Final Review

**REAL-TIME CROWD-AWARE SMART
ENVIRONMENTAL AUTOMATION SYSTEM**

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Problem Statement

"Bridging the gap between crowd awareness and environmental control remains a critical challenge in modern smart infrastructure."

- Commercial buildings account for nearly 40% of global carbon emissions.
- Existing systems use fixed schedules and basic occupancy sensors, limiting adaptability.
- Current solutions lack a real-time link between crowd density and energy demand.
- Inefficient control causes 20–30% unnecessary energy wastage.
- This leads to higher costs, reduced comfort, and poor environmental efficiency.
- An intelligent crowd-aware automation system is needed for dynamic environmental control.

Objective's

"To create an intelligent, data-driven automation framework that integrates crowd analytics and environmental sensing for adaptive and energy-efficient building management."

- Develop a real-time crowd-aware environmental automation system using AI, IoT, and Edge Computing.
- Detect, track, and analyze crowd density from live video streams.
- Continuously monitor temperature and humidity using IoT sensors.
- Dynamically control HVAC, lighting, and ventilation based on crowd conditions.
- Enable low-latency decision-making through Raspberry Pi-based edge deployment.
- Improve energy efficiency, occupant comfort, and operational sustainability.

Sustainable Development Goals - Mapping



- **SDG 7 – Affordable and Clean Energy:** 28.5% energy reduction through intelligent crowd-aware HVAC/lighting control.
- **SDG 11 – Sustainable Cities and Communities:** Smart building automation for commercial spaces, public venues, and urban infrastructure.
- **SDG 13 – Climate Action:** Reducing the ~40% carbon emissions attributed to commercial buildings via intelligent automation.

Literature Review

"Our literature survey helped identify suitable datasets and deep learning models for crowd analysis. Based on the survey, we selected the ShanghaiTech dataset, YOLOv8 for person detection, and a ResNet-based model for crowd density estimation."

- Surveyed recent research papers on crowd counting and density estimation using Computer Vision and Deep Learning.
- Compared publicly available datasets and selected the ShanghaiTech Dataset because of its diverse crowd scenes and ground-truth annotations.
- Studied YOLOv8 for real-time human detection and selected a pretrained model from Hugging Face.
- Reviewed ResNet-based crowd density estimation techniques and successfully trained the model using the ShanghaiTech dataset.
- Identified the next phase of work as integrating YOLOv8, Density Estimation, DeepSORT, and IoT-based environmental automation.

Existing Technology.

"Existing crowd monitoring systems primarily rely on computer vision, deep learning, and IoT technologies. However, most solutions focus on monitoring rather than intelligent environmental automation."

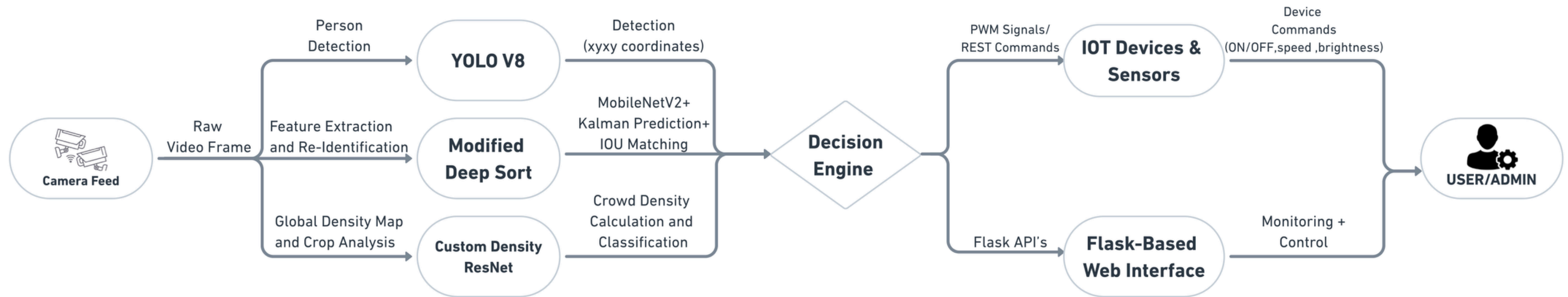
- Fixed-schedule HVAC and rule-based building automation are widely used.
- PIR sensors provide only basic occupancy detection.
- Environmental controls lack awareness of real-time crowd density.
- Existing systems face challenges in scalability, latency, and adaptability.
- Computer Vision and IoT have improved monitoring and energy management.
- Edge AI and Deep Learning are driving next-generation smart automation.

Proposed Work

"The methodology establishes a unified pipeline that transforms crowd intelligence and environmental sensing into automated, energy-efficient control decisions."

- Capture real-time video and environmental data through cameras and IoT sensors.
- Detect and track individuals using YOLOv8 and DeepSORT.
- Analyze crowd density using a deep learning-based estimation framework.
- Process crowd and sensor data on Raspberry Pi edge hardware.
- Generate adaptive control actions for HVAC, lighting, and ventilation.
- Visualize system status through a web-based monitoring interface.

System Architecture



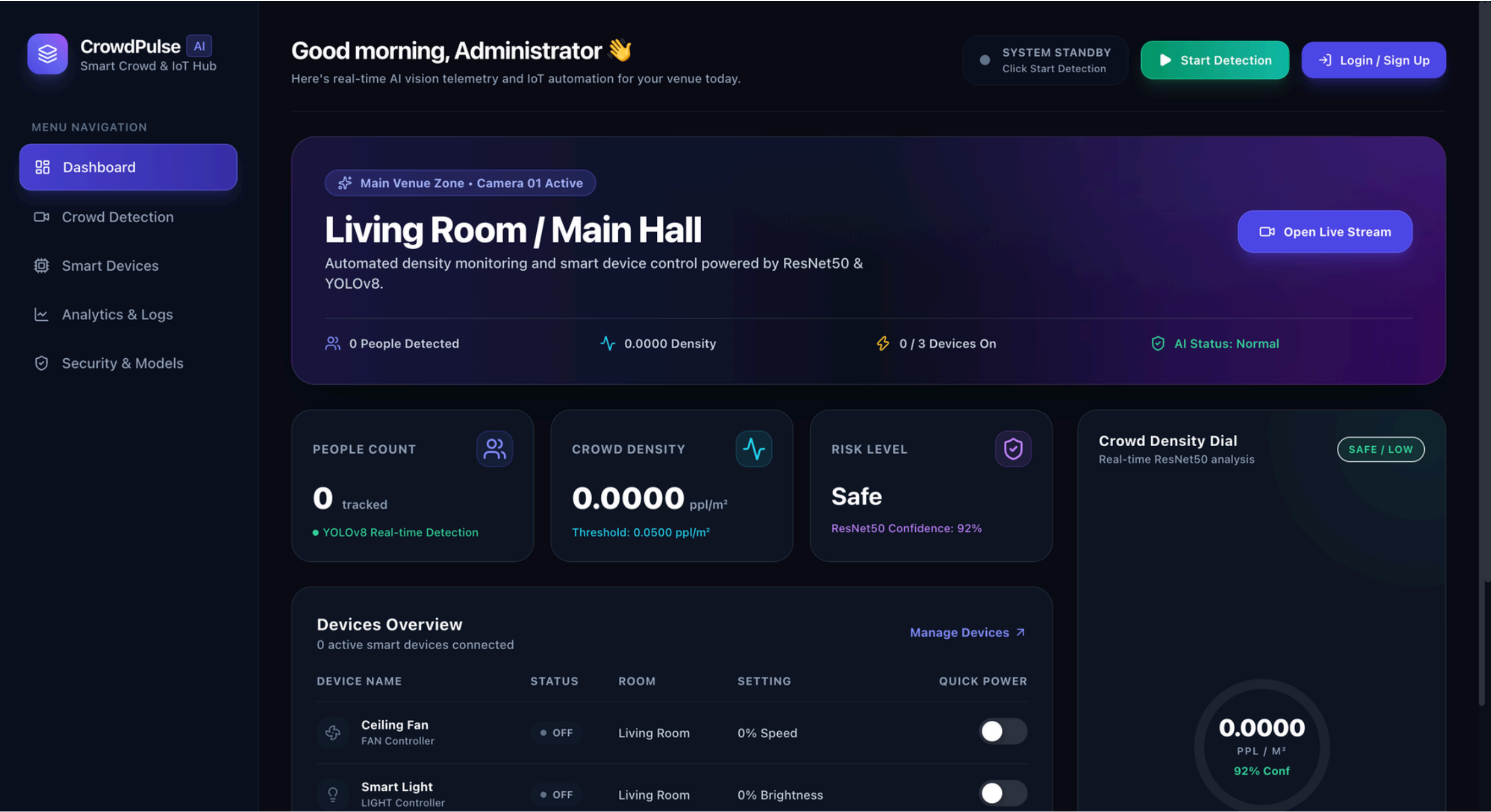
“Workflow of the Project (both Software and Hardware Integration)”

Expected Outcomes

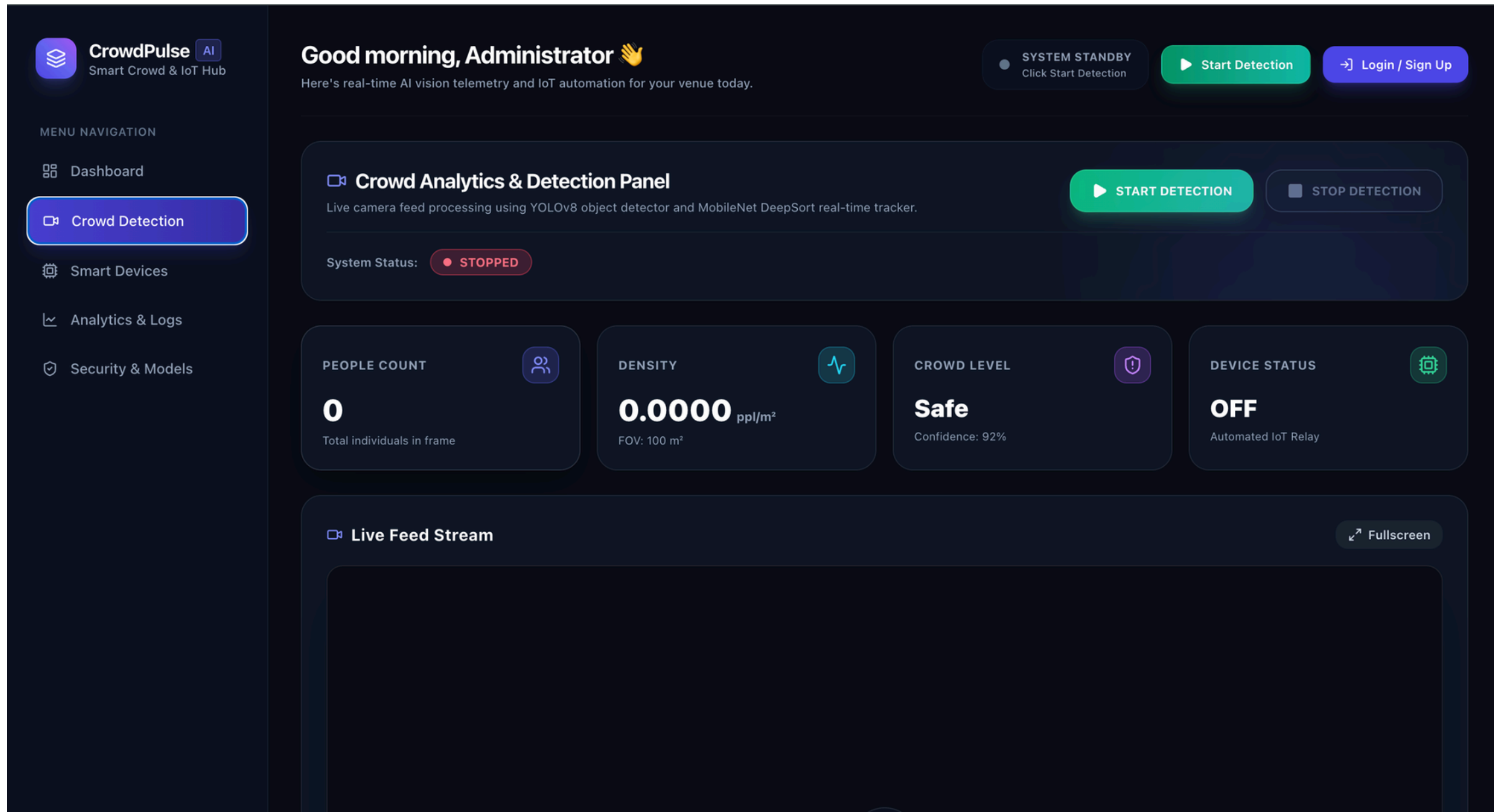
"The proposed system aims to deliver a scalable, energy-efficient, and intelligent automation framework that adapts environmental conditions to real-time crowd dynamics."

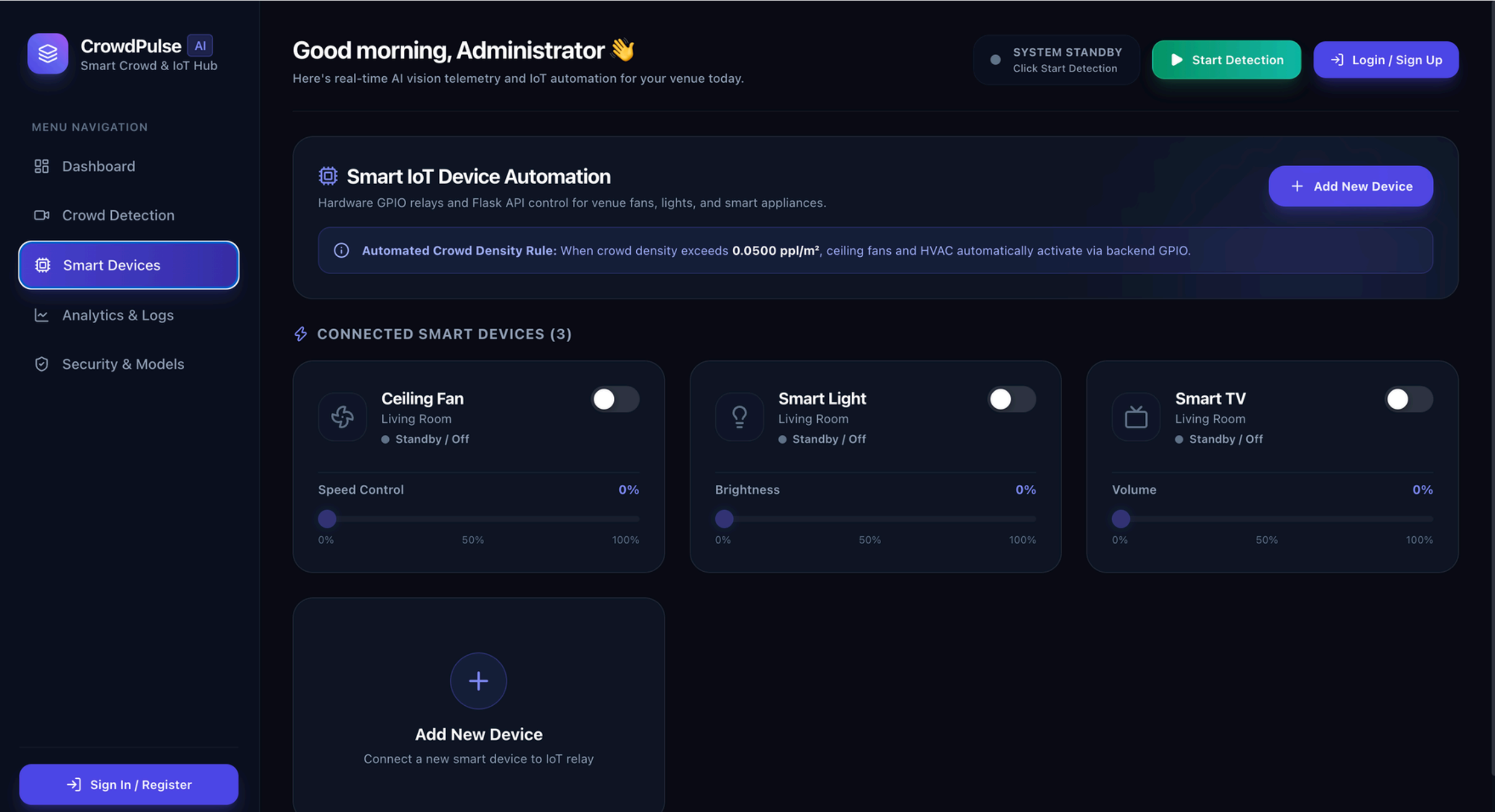
- Enable accurate real-time crowd detection, tracking, and density estimation.
- Achieve adaptive control of HVAC, lighting, and ventilation based on crowd conditions.
- Improve energy efficiency through intelligent environmental automation.
- Enhance occupant comfort by maintaining optimal environmental conditions.
- Provide real-time monitoring and analytics through a centralized dashboard.
- Support deployment across smart buildings, campuses, transportation hubs, and public spaces.

Dashboard



Crowd detection page





Time frame/ Schedule for completion

Task	Week 1	Week 2	Week 3	Week 4
Literature Review & Problem Scoping				
Dataset Collection & Preprocessing				
YOLOv8 Model Training & Tuning				
DensityResNet Architecture & Training				
DeepSORT Integration & Validation				
IoT Sensor Setup (DHT11 + Pi)				
Decision Engine Development				
Flask Web Interface Development				
System Integration & Testing				
Performance Evaluation & Ablation				
Documentation & Report Preparation				

Expecting Expense and Budget

- **Raspberry Pi 4 Model B (8 GB) — ₹ 4000**
- **DHT11 Temperature & Humidity Sensors — ₹ 350**
- **Hardware:** Raspberry Pi 4 (8 GB), DHT11 sensors, camera, LED arrays, fan modules, relay switches, breadboard, jumper wires, 5V power supply
- **Software & Frameworks:** Python, PyTorch, OpenCV, Ultralytics YOLOv8, Flask, TensorFlow Lite

References

No.	Title	Authors
1	Analysis of People Flow Image Detection System Based on Computer Vision Sensor - 2022	Bing Ou, Jingjing Yang, Wei Wang
2	A Computer Vision-Based Occupancy and Equipment Usage Detection Approach for Reducing Building Energy Demand - 2021	Paige Wenbin Tien, Shuangyu Wei, John Calautit
3	Enhancing Energy Efficiency in Commercial Office Buildings: A Smart IoT and Machine Vision Approach - 2024	Sushrut Madhav Patankar, Raju Kumar Swami, Sameer Suresh Nanivadekar
4	Intelligent Automation System for Energy Conservation Using IoT and Machine Learning - 2023	B. Sakthi Kumar, S. Ramalingam, S. Viveka, V. G. Vigneshwar, N. Sivasubramaniam, J. Sancta Maria
5	Low-Cost Image-Based Occupancy Sensor Using Deep Learning - 2021	T. M. Sanjeev Kumar, Susan G. Varghese, Ciji Pearl Kurian, Mouli Chandra Mouli



THANK YOU